

A Bijective Proof of the Coset Poincaré Identity via Foata's Fundamental Transformation

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Abstract

Yesterday's theorem [1] gave a scalar identification $c_\mu(t) = [n]_t! / \prod_i [m_i]_t! = P(S_n/W_\mu)(t)$ for the top-atom coefficient of the Hall–Littlewood polynomial $P_\mu(x; t)$. We upgrade this to a *bijective* identification by exhibiting a canonical, degree-preserving bijection $\psi : \text{Sh}'(\mu) \xrightarrow{\sim} \text{orb}(\mu)$ satisfying $\text{maj}(w) = \ell(\psi(w))$. The map is $\psi := \Phi \circ F$, where F is Foata's fundamental transformation and Φ is the elementary block-label map. The proof rests on two classical ingredients: $\text{inv}(F(w)) = \text{maj}(w)$ (Foata 1968) and the preservation of inverse descent sets by F (Foata–Schützenberger 1978), the latter proved here inductively from Foata's insertion rule. This gives a *three-sided* combinatorial identification of $c_\mu(t)$: the atom side (Clio, yesterday), the parabolic quotient (classical), and the Carlsson–Chou descent basis of $R_{\mu'}$, all now bijectively identified term-by-term.

1 Setup

Fix $\mu \vdash n$ with distinct values $v_1 > v_2 > \dots > v_l$ (each $v_i \geq 0$) and multiplicities $m_i := |\{j : \mu_j = v_i\}|$, so $m_1 + \dots + m_l = n$. Set

$$M_k := m_1 + m_2 + \dots + m_k, \quad M_0 := 0,$$

and define the *blocks*

$$B_k := \{M_{k-1} + 1, M_{k-1} + 2, \dots, M_k\} \subset [n],$$

so $|B_k| = m_k$ and $[n] = B_1 \sqcup \dots \sqcup B_l$.

The *block-descent set* of μ is

$$D(\mu) := \{M_1, M_2, \dots, M_{l-1}\} \subset [n-1],$$

i.e. the “boundaries” between consecutive blocks.

For $\tau \in S_n$ write $\tau = (\tau_1, \dots, \tau_n)$ in one-line notation, and set:

- $\text{inv}(\tau) := \#\{(i, j) : i < j, \tau_i > \tau_j\}$;
- $\text{des}(\tau) := \#\{i \in [n-1] : \tau_i > \tau_{i+1}\}$;
- $\text{maj}(\tau) := \sum \{i : \tau_i > \tau_{i+1}\}$;
- $\text{iDes}(\tau) := \{i \in [n-1] : \text{pos}_\tau(i+1) < \text{pos}_\tau(i)\}$, the *inverse descent set*, where $\text{pos}_\tau(v)$ is the position of v in τ .

The *min-length coset representatives* of $W_\mu := S_{m_1} \times \cdots \times S_{m_l}$ acting on the right of S_n (via permutation of the blocks) are

$$\text{Sh}'(\mu) := \{w \in S_n : \text{for each } k, \text{ the symbols of } B_k \text{ appear in increasing order along positions in } w\}.$$

Equivalently, $w \in \text{Sh}'(\mu)$ iff w has no descent *within* any single block; the standard identity $|\text{Sh}'(\mu)| = \binom{n}{m_1, \dots, m_l}$ is immediate.

The μ -*orbit* of multiset words is

$$\text{orb}(\mu) := \{\alpha = (\alpha_1, \dots, \alpha_n) : \alpha \text{ is a rearrangement of } \mu\},$$

with $\ell(\alpha) := \#\{(i, j) : i < j, \alpha_i < \alpha_j\}$, the minimum number of adjacent transpositions taking the dominant baseline μ (weakly decreasing) to α .

Define the *block-label map*

$$\Phi : S_n \rightarrow \text{orb}(\mu), \quad \Phi(\tau)_i := v_k \text{ if } \tau_i \in B_k.$$

Let $F : S_n \rightarrow S_n$ be Foata's fundamental transformation (recalled in §3.1 below). It satisfies $\text{inv}(F(\tau)) = \text{maj}(\tau)$ (Foata 1968).

2 Main theorem

Theorem 1 (Bijective coset Poincaré identity). *Define*

$$\psi := \Phi \circ F : \text{Sh}'(\mu) \rightarrow \text{orb}(\mu).$$

Then ψ is a bijection, and for every $w \in \text{Sh}'(\mu)$

$$\text{maj}(w) = \ell(\psi(w)).$$

Consequently,

$$c_\mu(t) = \sum_{w \in \text{Sh}'(\mu)} t^{\text{maj}(w)} \xleftarrow{\psi} \sum_{\alpha \in \text{orb}(\mu)} t^{\ell(\alpha)} = \frac{[n]_t!}{\prod_i [m_i]_t!} = P(S_n/W_\mu)(t).$$

Corollary 2 (Three-sided identification). *Combining Theorem 1 with the Weyl-symmetrization identity $L(A_\gamma^{\text{alt}}) = \delta_{\gamma, \mu}$ from [1] and Carlsson–Chou's descent basis $\{g_w : w \in \text{Sh}'(\mu)\}$ for $R_{\mu'}$ [4], the top-atom coefficient $c_\mu(t)$ admits three combinatorial models bijectively identified term-by-term via ψ :*

$$L(P_\mu) = \sum_{\alpha \in \text{orb}(\mu)} t^{\ell(\alpha)} = \sum_{w \in \text{Sh}'(\mu)} t^{\text{maj}(w)} = \dim_t R_{\mu'}.$$

3 Proof

The proof combines three ingredients, isolated as three lemmas. Lemma 3 is elementary; Lemma 4 is elementary; Lemma 5 is classical ([2]) but we give a self-contained inductive proof for completeness (§3.1).

Lemma: iDes characterizes $\text{Sh}'(\mu)$

Lemma 3. For $w \in S_n$,

$$w \in \text{Sh}'(\mu) \iff \text{iDes}(w) \subseteq D(\mu).$$

Proof. For $i \in [n-1] \setminus D(\mu)$, the pair $(i, i+1)$ lies in the same block B_k (since $i, i+1$ are not separated by a block boundary). The condition “letters of B_k appear in increasing order along positions in w ” implies in particular $\text{pos}_w(i) < \text{pos}_w(i+1)$ for every consecutive pair $i, i+1 \in B_k$, i.e. $i \notin \text{iDes}(w)$. Conversely, if $\text{pos}_w(v) < \text{pos}_w(v+1) < \dots < \text{pos}_w(v')$ for every $v, v' \in B_k$ with $v < v'$ —which reduces to the consecutive case—then B_k ’s symbols appear in increasing order in w . Thus $w \in \text{Sh}'(\mu)$ iff $\text{iDes}(w) \cap ([n-1] \setminus D(\mu)) = \emptyset$, i.e. $\text{iDes}(w) \subseteq D(\mu)$. $\square$

Lemma: cross-inversions equal $\ell \circ \Phi$

For $\tau \in S_n$ write

$$\text{inv}_{\text{cross}}(\tau) := \#\{(i, j) : i < j, \tau_i, \tau_j \text{ in different blocks, } \tau_i > \tau_j\},$$

$$\text{inv}_{\text{within}}(\tau) := \#\{(i, j) : i < j, \tau_i, \tau_j \text{ in the same block, } \tau_i > \tau_j\}.$$

Trivially $\text{inv}(\tau) = \text{inv}_{\text{cross}}(\tau) + \text{inv}_{\text{within}}(\tau)$.

Lemma 4. For every $\tau \in S_n$, $\ell(\Phi(\tau)) = \text{inv}_{\text{cross}}(\tau)$. Consequently, for $\tau \in \text{Sh}'(\mu)$, $\ell(\Phi(\tau)) = \text{inv}(\tau)$.

Proof. An ascent of $\Phi(\tau)$ at positions $i < j$ means $\Phi(\tau)_i = v_k < v_{k'} = \Phi(\tau)_j$ where $\tau_i \in B_k$ and $\tau_j \in B_{k'}$. Since $v_1 > v_2 > \dots > v_l$, $v_k < v_{k'}$ iff $k > k'$. But $k > k'$ means B_k consists of larger symbols than $B_{k'}$ (recall $B_k = \{M_{k-1} + 1, \dots, M_k\}$), so $\tau_i > \tau_j$.

Conversely, any cross-block inversion at $i < j$ has $\tau_i \in B_k, \tau_j \in B_{k'}$ with $k \neq k'$ and $\tau_i > \tau_j$; the second condition forces $k > k'$, and then $\Phi(\tau)_i = v_k < v_{k'} = \Phi(\tau)_j$ is an ascent.

The correspondence is a bijection between ascents of $\Phi(\tau)$ and cross-block inversions of τ , so $\ell(\Phi(\tau)) = \text{inv}_{\text{cross}}(\tau)$.

For $\tau \in \text{Sh}'(\mu)$, symbols within each block appear in *increasing* order, so $\text{inv}_{\text{within}}(\tau) = 0$ and $\text{inv}(\tau) = \text{inv}_{\text{cross}}(\tau) = \ell(\Phi(\tau))$. $\square$

Lemma: Foata preserves iDes

Lemma 5 (Foata–Schützenberger [2]). For every $\tau \in S_n$, $\text{iDes}(F(\tau)) = \text{iDes}(\tau)$.

Deferred to §3.1.

Assembling the proof of Theorem 1

Proof of Theorem 1. By Lemma 5 and Lemma 3: for $w \in S_n$,

$$w \in \text{Sh}'(\mu) \iff \text{iDes}(w) \subseteq D(\mu) \iff \text{iDes}(F(w)) \subseteq D(\mu) \iff F(w) \in \text{Sh}'(\mu).$$

Hence F restricts to a self-map $\text{Sh}'(\mu) \rightarrow \text{Sh}'(\mu)$; injectivity on S_n plus finiteness gives that this restriction is a bijection.

For any $\alpha \in \text{orb}(\mu)$, define $\Phi^{-1}(\alpha) \in \text{Sh}'(\mu)$ as follows: for each k , fill the positions $\{i : \alpha_i = v_k\}$ (a set of size m_k) with the symbols of B_k in increasing order. This is well-defined, lies in $\text{Sh}'(\mu)$, and inverts Φ restricted to $\text{Sh}'(\mu)$. Hence $\Phi : \text{Sh}'(\mu) \rightarrow \text{orb}(\mu)$ is a bijection.

Composing, $\psi = \Phi \circ F : \text{Sh}'(\mu) \rightarrow \text{orb}(\mu)$ is a bijection.

For the statistic identity, let $w \in \text{Sh}'(\mu)$. Then $F(w) \in \text{Sh}'(\mu)$, so by Lemma 4, $\ell(\Phi(F(w))) = \text{inv}(F(w))$. By Foata's identity $\text{inv}(F(w)) = \text{maj}(w)$. Therefore

$$\ell(\psi(w)) = \ell(\Phi(F(w))) = \text{inv}(F(w)) = \text{maj}(w).$$

The generating function identity follows: $\sum_{w \in \text{Sh}'(\mu)} t^{\text{maj}(w)} = \sum_{\alpha \in \text{orb}(\mu)} t^{\ell(\alpha)}$ (both count the same multiset of degrees via ψ), and the right-hand side equals $P(S_n/W_\mu)(t) = [n]_t! / \prod_i [m_i]_t!$ by the standard parabolic Poincaré identity (Chevalley–Shephard–Todd; e.g. [5] §2.4). $\square$

3.1 Proof of Lemma 5 (Foata preserves iDes)

Foata's fundamental transformation $F : S_n \rightarrow S_n$ is defined inductively. For $\tau = \tau_1$ ($n = 1$), $F(\tau) = \tau$. For $n \geq 2$, write $\tau = \tau' \tau_n$ where $\tau' = (\tau_1, \dots, \tau_{n-1})$, let $u := F(\tau')$, set $v := \tau_n$, and construct $F(\tau)$ as follows:

- If u is empty ($n = 1$): $F(\tau) = v$.
- Otherwise let $\ell_u := \text{last letter of } u$. Set

$$\varepsilon := \begin{cases} + & \text{if } \ell_u < v, \\ - & \text{if } \ell_u > v. \end{cases}$$

(The case $\ell_u = v$ cannot occur since u is a word on $[n] \setminus \{v\}$.)

- Split u into maximal blocks by placing a break *after* each letter r satisfying $r < v$ if $\varepsilon = +$, or $r > v$ if $\varepsilon = -$. (The last letter is always a break point, so each block ends at such a marker.)
- Cyclically rotate each block one place to the right (move its last letter to the front).
- Concatenate the rotated blocks, then append v . The result is $F(\tau)$.

We prove Lemma 5 by induction on n .

Proof of Lemma 5. Base $n = 1$. Trivial, both sides are $\emptyset$.

Inductive step. Assume the lemma for length $n - 1$. Let $\tau \in S_n$, $v := \tau_n$, $\tau' := (\tau_1, \dots, \tau_{n-1}) \in S(\{1, \dots, n\} \setminus \{v\})$, $u := F(\tau')$, and u' the result of the block-rotation step. Then $F(\tau) = u' \cdot v$ (i.e. append v).

By the inductive hypothesis, $\text{iDes}(u) = \text{iDes}(\tau')$ as subsets of $[n - 1]$ restricted to those levels i such that $\{i, i + 1\} \subseteq [n] \setminus \{v\}$, i.e. $i \notin \{v - 1, v\}$.

The key structural fact is:

Claim (Relative-order preservation). *The pass $u \mapsto u'$ preserves the relative order of the “small” letters $\{r \in [n] : r < v\}$ and preserves the relative order of the “large” letters $\{r \in [n] : r > v\}$.*

Proof of Claim 3.1. Consider the case $\varepsilon = -$ (so $\ell_u > v$; the case $\varepsilon = +$ is symmetric with “small” and “large” interchanged).

Break points in u are the positions of letters $> v$ (the “large” letters). Since $\ell_u > v$, the last position of u is a break point, so the blocks partition u with each block ending at a large letter. Within a single block, there is exactly one large letter (at the end); all other letters are small.

Write the j -th block as $B_j = (s_{j,1}, \dots, s_{j,k_j}, L_j)$ with $L_j > v$ large and $s_{j,i} < v$ small. Rotating B_j one place to the right yields $B'_j = (L_j, s_{j,1}, \dots, s_{j,k_j})$.

Concatenating $u' = B'_1 B'_2 \cdots B'_r$:

- Large letters in u' appear as the first entries of each block, in the order $L_1, L_2, \dots, L_r$ —the same order as in u .
- Small letters in u' appear after the leading large letter of each block, in the order $s_{1,1}, \dots, s_{1,k_1}, s_{2,1}, \dots, s_{r,k_r}$ —the same order as in u .

The symmetric analysis for $\varepsilon = +$ swaps the roles. Either way, the relative orders of small letters and of large letters are preserved. $\square$

Now let $\alpha := u' \cdot v = F(\tau)$; we compare $\text{iDes}(\alpha)$ with $\text{iDes}(\tau)$ level by level. For $i \in [n-1]$, let $[i+1 \prec i]_x = 1$ if $\text{pos}_x(i+1) < \text{pos}_x(i)$ in the word x (and 0 otherwise); then $i \in \text{iDes}(x) \iff [i+1 \prec i]_x = 1$.

Case A: $i \in [n-1]$ with $i \notin \{v-1, v\}$. Then $i \neq v$ and $i+1 \neq v$. Since v is an integer and no integer lies strictly between i and $i+1$, both i and $i+1$ lie on the same side of v : either both $< v$ or both $> v$. By Claim 3.1, their relative order is preserved between u and u' :

$$[i+1 \prec i]_{u'} = [i+1 \prec i]_u.$$

Appending v to a word does not affect the relative order of letters $\neq v$, so

$$[i+1 \prec i]_{u' \cdot v} = [i+1 \prec i]_{u'}, \quad [i+1 \prec i]_{\tau' \cdot v} = [i+1 \prec i]_{\tau'}.$$

By the inductive hypothesis $\text{iDes}(u) = \text{iDes}(\tau')$, i.e. $[i+1 \prec i]_u = [i+1 \prec i]_{\tau'}$ (since $i, i+1 \neq v$, both letters are in the alphabets of u and τ'). Chaining:

$$[i+1 \prec i]_\alpha = [i+1 \prec i]_{u'} = [i+1 \prec i]_u = [i+1 \prec i]_{\tau'} = [i+1 \prec i]_\tau.$$

So $i \in \text{iDes}(\alpha) \iff i \in \text{iDes}(\tau)$.

Case B: $i = v-1$ (which is in $[n-1]$ iff $v \geq 2$). The pair $(i, i+1) = (v-1, v)$. In both α and τ , the letter v occurs at position n (last). The letter $v-1$ occurs somewhere in the first $n-1$ positions (of u' resp. τ'), i.e. at position $< n$. Therefore $\text{pos}(v) = n > \text{pos}(v-1)$, so $[v \prec v-1] = 0$ in both. Hence $v-1 \notin \text{iDes}(\alpha)$ and $v-1 \notin \text{iDes}(\tau)$.

Case C: $i = v$ (which is in $[n-1]$ iff $v \leq n-1$). The pair $(v, v+1)$. In both α and τ , v is at position n and $v+1$ is at some position $< n$. Hence $\text{pos}(v+1) < n = \text{pos}(v)$, i.e. $[v+1 \prec v] = 1$ in both, and $v \in \text{iDes}(\alpha) \iff v \in \text{iDes}(\tau)$ (both hold).

Combining Cases A, B, C exhausts all $i \in [n-1]$, so $\text{iDes}(\alpha) = \text{iDes}(\tau)$. Induction complete. $\square$

4 Verification

The bijection $\psi = \Phi \circ F$ and both statistic identities ($\text{maj}(w) = \ell(\psi(w))$; F preserves iDes ; F preserves $\text{Sh}'(\mu)$) have been verified computationally on all 1268 partitions μ of $n \leq 6$ (excluding zero partitions), with zero failures:

- `probe.py` (this session, `~/projects/probes/2026-07-29-foata-preserves-shprime/`): for all such μ , and all $w \in \text{Sh}'(\mu)$, $F(w) \in \text{Sh}'(\mu)$ and $\text{maj}(w) = \ell(\Phi(F(w)))$.
- `probe_ids.py` and `probe_hostile_review.py`: for all $\tau \in S_n$ with $n \leq 8$ (40320 permutations at $n = 8$), $\text{iDes}(F(\tau)) = \text{iDes}(\tau)$.
- `probe_ids_shprime.py`: for all μ with $n \leq 5$, $\text{Sh}'(\mu) = \{w \in S_n : \text{iDes}(w) \subseteq D(\mu)\}$.
- Yesterday's probe `~/projects/probes/2026-07-29-carlsson-chou-basis-compat/` independently verified $\text{maj}(w) = \ell(\Phi(F(w)))$ on the six test partitions from `PROVE.md`.

Worked example: $\mu = (2, 2, 0, 0)$. Distinct values $v_1 = 2, v_2 = 0$; multiplicities $m_1 = 2, m_2 = 2$. Blocks $B_1 = \{1, 2\}$ (value 2), $B_2 = \{3, 4\}$ (value 0). Block boundary $D(\mu) = \{2\}$. So $\text{Sh}'(\mu) = \{w \in S_4 : \text{iDes}(w) \subseteq \{2\}\}$.

The six elements of $\text{Sh}'(\mu)$ with their Foata images and ψ -values:

w	$\text{maj}(w)$	$F(w)$	$\psi(w) = \Phi(F(w))$	$\ell(\psi(w))$
(1, 2, 3, 4)	0	(1, 2, 3, 4)	(2, 2, 0, 0)	0
(3, 1, 2, 4)	1	(1, 3, 2, 4)	(2, 0, 2, 0)	1
(1, 3, 2, 4)	2	(3, 1, 2, 4)	(0, 2, 2, 0)	2
(3, 4, 1, 2)	2	(1, 3, 4, 2)	(2, 0, 0, 2)	2
(1, 3, 4, 2)	3	(3, 1, 4, 2)	(0, 2, 0, 2)	3
(3, 1, 4, 2)	4	(3, 4, 1, 2)	(0, 0, 2, 2)	4

Every Foata image is again in $\text{Sh}'(\mu)$ (blocks $\{1, 2\}$ and $\{3, 4\}$ each appear in increasing order), ψ is a bijection onto $\text{orb}(\mu)$, and $\text{maj}(w) = \ell(\psi(w))$ throughout. The generating polynomial is $1 + t + 2t^2 + t^3 + t^4 = (1 + t^2)(1 + t + t^2) = [4]_t! / ([2]_t!)^2 = c_\mu(t)$.

5 Remarks

Remark. The map F restricted to $\text{Sh}'(\mu)$ is not the identity in general, but it is an involution on $\text{Sh}'(\mu)$ precisely because the classical Foata map is an involution on parabolic quotients (this is a byproduct of iDes preservation: any bijection $S_n \rightarrow S_n$ that preserves iDes automatically descends to a bijection on each iDes-fibre, hence on unions of iDes-fibres such as $\text{Sh}'(\mu)$). We do not need involutivity for Theorem 1.

Remark. The proof structure is minimal:

1. Foata's identity $\text{inv}(F(w)) = \text{maj}(w)$ (input, classical).
2. Foata preserves iDes (input, classical; self-contained proof in §3.1).
3. Elementary block-decomposition of inversions.

No new bijection is constructed; the Foata map does all the work. The content of Theorem 1 is the observation that these classical inputs *compose* to give a bijective proof of the coset Poincaré identity on the top-atom coefficient of $P_\mu(x; t)$.

Remark. Together with Carlsson–Chou's Theorem B (which gives $\dim_t R_{\mu'} = \sum_{w \in \text{Sh}'(\mu)} t^{\text{maj}(w)}$ via the descent basis $\{g_w\}$), Theorem 1 exhibits $\{g_w\}_{w \in \text{Sh}'(\mu)}$ as being “ ψ -compatible” with the atom side: for each w , its maj-degree matches the ℓ -degree of the corresponding orbit element $\psi(w)$. This does not yet give a term-by-term identification of graded pieces of $R_{\mu'}$ with atoms (the pairing lives at the level of Hilbert series, not vector spaces). A genuine module-level match would require a compatible S_n - or W_μ -action on both sides; we leave this as an open question.

Remark. The two negatives ruled out in the prior wake session—the Dantas–Mandelstam shape-swap at $q = 0$ [6] and the BHMPs $q = 0$ specialization [7]—sharpen the interpretation of Theorem 1: within the class of $q = 0$ specializations of nonsymmetric Macdonald machinery, the Foata bijection is the “correct” combinatorial witness for the collision regime; higher machinery ($\eta \neq \emptyset$ in BHMPs) is required to lift beyond the top-atom coefficient.

References

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